

Bangalore Zomato Restaurant Analysis Using Python, PostgreSQL and Power BI

Report Structure

Cover Page

Project Title: Bangalore Zomato Restaurant Analysis

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Date: [Dataset Link Attach](#)

>Abstract

The **Bangalore Zomato Restaurant Analysis** project aims to analyze restaurant data from Bangalore to identify customer preferences, restaurant distribution, pricing trends, ratings, and business opportunities. The restaurant industry generates large volumes of data every day, making data analysis essential for understanding customer behavior and supporting better business decisions.

The project began with data preprocessing using **Python (Pandas)**, where the raw dataset was cleaned by handling missing values, removing duplicate records, converting data types, and preparing the data for further analysis. The cleaned dataset was then imported into **PostgreSQL**, where SQL queries were used to perform business analysis, including identifying the most popular restaurant locations, top cuisines, restaurant types, average ratings, average costs, customer voting patterns, online ordering trends, and table booking statistics.

An interactive dashboard was developed using **Power BI** to present the analytical results through KPI cards, charts, and slicers. The dashboard enables users to explore restaurant performance,

compare locations, analyze cuisine popularity, and understand customer preferences in an easy-to-use visual format.

The analysis revealed that **BTM, Whitefield,** and **HSR** contain the highest concentration of restaurants, while **North Indian cuisine** and **Casual Dining** restaurants dominate the Bangalore food market. Most restaurants provide online ordering services, whereas table booking facilities are comparatively limited. These insights can help restaurant owners, business managers, and decision-makers improve customer satisfaction, optimize pricing strategies, and expand operations in high-demand locations.

Overall, this project demonstrates the complete data analytics workflow—from data cleaning and SQL analysis to interactive dashboard development—using **Python, PostgreSQL, and Power BI**. It highlights the practical application of data analytics techniques for solving real-world business problems and generating actionable insights.

>Introduction

- **What is Zomato?**

Zomato is a prominent Indian food-tech platform that connects consumers, restaurant partners, and delivery personnel

- **Importance of restaurant analytics**

Restaurant analytics transforms raw operational and customer data into actionable insights, [helping operators optimize costs, improve profit margins, and personalize the dining experience](#). In a highly competitive industry, leveraging this data is critical to tracking performance, managing resources effectively, and staying ahead of market trends

- **Why data analysis is important in the food industry**

Data analysis is crucial in the food industry because [it transforms raw information into actionable insights, driving operational efficiency, stricter food safety, and personalized customer experiences](#). It allows businesses to minimize waste, predict market trends, and guarantee consistent product quality from farm to table

- **Why Bangalore was selected**

Bengaluru was selected for Zomato data analysis [because it is India's IT capital with a highly concentrated, fast-paced migrant population that heavily relies on dining out and food delivery](#). With over 12,000 restaurants, the city's diverse and rapidly growing food scene provides the perfect, large-scale dataset to study

>Problem Statement

Restaurant owners receive huge amounts of customer and operational data but often fail to convert it into business decisions.

This project aims to identify:

- Popular restaurant locations
- Popular cuisines
- Customer preferences
- Cost distribution
- Restaurant ratings
- Online ordering trends
- Table booking trends

>Objectives

- Clean raw restaurant data
- Remove duplicate records
- Handle missing values

- Analyse restaurant performance
- Build interactive dashboards
- Generate business recommendations

>Dataset Description

Source: Kaggle

Dataset Size

- Original Records: 51,717
- Cleaned Records: 19,924
- Number of Columns: 11

Column Description

Column	Description
Name	Restaurant Name
Location	Area
Rate	Restaurant Rating
Votes	Customer Votes
Online Order	Yes/No
Book Table	Yes/No
Rest Type	Restaurant Category
Cuisines	Food Type
Approx Cost	Average Cost
Listed In	Restaurant Category
City	Bangalore Area

>Tools Used

Python

- Pandas
- NumPy
- Matplotlib

PostgreSQL

Power BI

Jupyter Notebook

>Data Cleaning (Python)

1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Explanation: Imported libraries for data manipulation and visualization.

2. Load Dataset

```
df = pd.read_csv("zomota123.csv")
df.head()
```

Explanation: Loaded the dataset into a Pandas DataFrame.

3. Select Important Columns

```
clean_df = df[[
    "name",
    "online_order",
    "book_table",
    "rate",
```

```
"votes",  
"location",  
"rest_type",  
"cuisines",  
"approx_cost(for two people)",  
"listed_in(type)",  
"listed_in(city)"  
]]
```

Explanation: Selected only the relevant columns required for analysis.

4. Check Missing Values

```
clean_df.isnull().sum()
```

Explanation: Identified missing values before data cleaning.

5. Remove Missing Values

```
clean_df = clean_df.dropna()
```

Explanation: Removed rows containing null values.

6. Remove Duplicate Records

```
clean_df = clean_df.drop_duplicates()
```

Explanation: Removed duplicate restaurant entries.

7. Convert Cost Column

```
clean_df["approx_cost(for two people)"] = (  
    clean_df["approx_cost(for two people)"]  
    .str.replace(",", "")  
    .astype(int)  
)
```

Explanation: Removed commas and converted the cost column into an integer.

8. Rename Columns

```
clean_df.rename(columns={  
    "approx_cost(for two people)": "Approx_cost"  
}, inplace=True)
```

Explanation: Renamed the column for easier use in SQL and Power BI.

9. Save Cleaned Dataset

```
clean_df.to_csv("clean_zomato.csv", index=False)
```

Explanation: Exported the cleaned dataset for PostgreSQL and Power BI.

10. Generate KPIs

```
print("Total Restaurants :", clean_df["name"].count())  
print("Total Locations :", clean_df["location"].nunique())  
print("Average Rating :", round(clean_df["rate"].mean(), 2))  
print("Average Cost :", round(clean_df["Approx_cost"].mean(), 2))  
print("Total Votes :", clean_df["votes"].sum())
```

Explanation: Calculated the key performance indicators used in the dashboard.

Top 10 Restaurant Locations

```
clean_df["location"].value_counts().head(10).plot(kind="bar")  
plt.title("Top 10 Restaurant Locations")  
plt.show()
```

Top 10 Restaurant Types

```
clean_df["rest_type"].value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Restaurant Types")
plt.show()
```

>SQL Analysis

Include SQL queries and outputs.

Examples:

1.Top 10 Locations

```
select "location",count(*)as total_restro from zam
group by "location"
order by total_restro desc
limit 10
```

	location text	total_restro bigint
1	BTM	1378
2	Whitefield	1151
3	HSR	1121
4	Indiranagar	1038
5	Marathahalli	956
6	JP Nagar	868
7	Koramangala 5th Blo...	836
8	Electronic City	758
9	Jayanagar	746
10	Bannerghatta Road	641

This query identifies the areas with the highest number of restaurants, helping businesses understand where restaurant density is greatest.

2.Top 10 Cuisines

```
select "cuisines",count(*)as total_restro from zam
group by "cuisines"
order by total_restro desc
limit 10
```

	cuisines text	total_restro bigint
1	North Indian	1157
2	North Indian, Chinese	827
3	South Indian	802
4	Biryani	366
5	Bakery, Desserts	348
6	Fast Food	287
7	South Indian, North Indian, Chin...	279
8	Desserts	243
9	Bakery	219
10	Chinese	190

This query identifies the Cuisines with the highest number of restaurants, helping businesses understand Restaurants Cuisines

3.Restaurant Types

```
select "rest_type",count(*)as total_restro from zam
group by "rest_type"
order by total_restro desc
limit 10
```

	rest_type text	total_restro bigint
1	Quick Bites	7162
2	Casual Dining	4248
3	Delivery	1246
4	Cafe	1114
5	Takeaway, Deliv...	984
6	Dessert Parlor	863
7	Casual Dining, B...	437
8	Bakery	407
9	Beverage Shop	348
10	Food Court	250

This query identifies the Total number of Restaurant type

4.Average Cost

```
select round(avg("Approx_cost"),1) as avg_cost from zam
```

	avg_cost numeric
1	417

This query calculates the average cost of restaurants

5.Average Rating

```
select round(avg(rate),1)as avg_rate from zam
```

	avg_rate numeric
1	3.7

This query counts average rating

6.Online Order Analysis

```
select "online_order", count(*) as restaurants
from zam
group by "online_order"
```

	online_order text	restaurants bigint
1	No	7878
2	Yes	12046

This Query Checks the Distribution of Online order

7.Table Booking Analysis

```
SELECT
    "book_table",
    COUNT(*) AS restaurants
FROM zam
GROUP BY "book_table";
```

	book_table text	restaurants bigint
1	No	17067
2	Yes	2857

This Query also checks the distribution of Booking Table

8.CTE

```
WITH location_count AS
(
    SELECT
        "location",
        COUNT(*) AS total_restaurants
    FROM zam
    GROUP BY "location"
)
```

```

SELECT *
FROM location_count
WHERE total_restaurants >
(
    SELECT AVG(total_restaurants)
    FROM location_count
)
ORDER BY total_restaurants DESC;

```

	location text	total_restaurants bigint
1	BTM	1378
2	Whitefield	1151
3	HSR	1121
4	Indiranagar	1038
5	Marathahalli	956
6	JP Nagar	868
7	Koramangala 5th Blo...	836
8	Electronic City	758
9	Jayanagar	746
10	Bannerghatta Road	641

This Query identifies number of restaurants in each location

9.CASE WHEN

```

SELECT
    "name",
    "rate",
    CASE
        WHEN "rate" >= 4.5 THEN 'Excellent'

```

```

    WHEN "rate" >= 4 THEN 'Very Good'

    WHEN "rate" >= 3 THEN 'Good'

    ELSE 'Average'

END AS rating_category

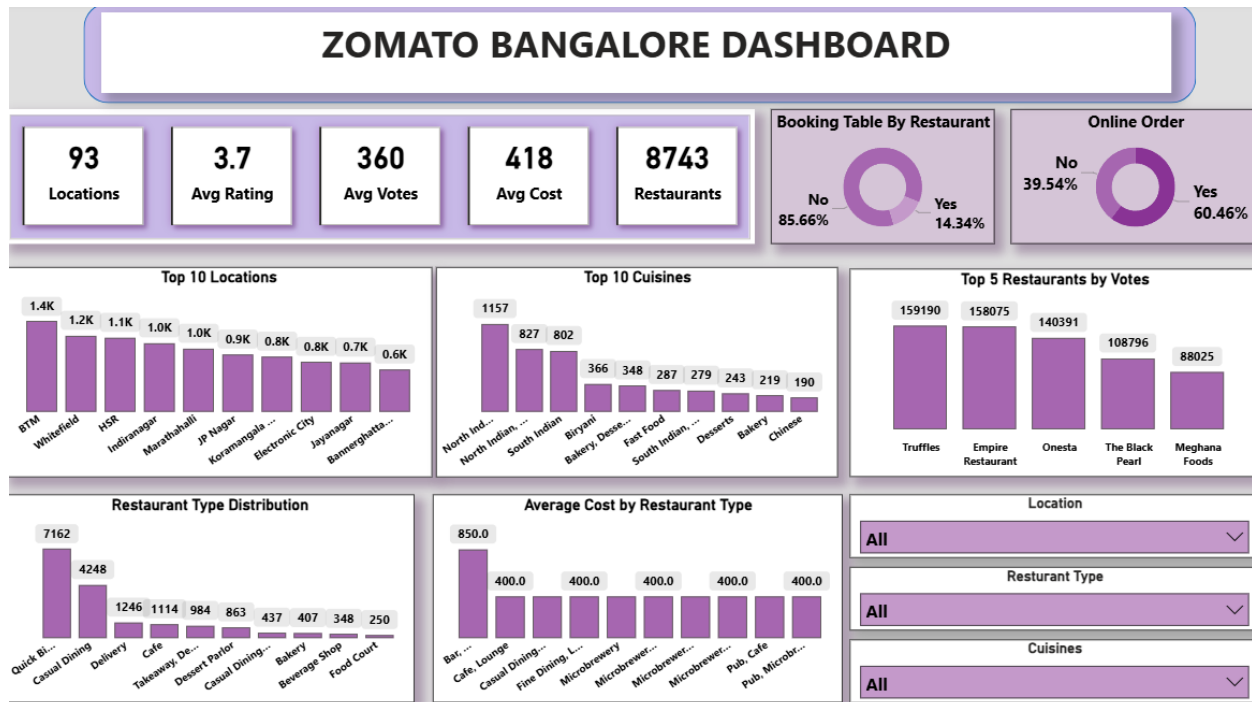
FROM zam;

```

	name text	rate numeric	rating_category text
1	Jalsa	4.1	Very Good
2	Spice Elephant	4.1	Very Good
3	San Churro Cafe	3.8	Good
4	Addhuri Udupi Bhojana	3.7	Good
5	Grand Village	3.8	Good
6	Timepass Dinner	3.8	Good
7	Rosewood International Hotel - Bar & Restaurant	3.6	Good
8	Onesta	4.6	Excellent
9	Penthouse Cafe	4	Very Good
10	Smaczego	4.2	Very Good

This query categorizes restaurants based on their ratings using the CASE statement.

>Power BI Dashboard



>Business Insights from Dashboard

- KPI (Key Performance Indicator)

Total location: 93

Average Rating: 3.7

Average Votes: 360

Average Cost: 418

Total Restaurants :8743

- In online order 60% of Restaurant taking order while 40% Restaurant taking offline order
- 14% of Restaurant booked table himself while 86% not booked table by himself
- BTM location Has most number of Restaurant and followed by Whitefield and HSR also have number of Restaurant.
- In Top 10 Cuisines North Indian Most Preferred Food in Restaurants, behind logic is that Bangalore is it hub Sector so many people coming for studies, job or startup form whole Indian or overseas also. Most of

People want roti-sabji, dal – chawl like daily food. In North Indian Food all included that why North Indian most number of preferred cuisines in restaurants.

- In Most Votes is truffle Restaurant
- Quick Bites is the most common restaurant type."
- In Average cost Bar and pubs are preferred

>Business Recommendations

- Expand restaurants in high-demand locations.
- Increase online ordering availability.
- Improve customer experience to increase ratings.
- Promote high-rated restaurants.
- Focus marketing on popular cuisines.

Challenges Faced

During the development of the Bangalore Zomato Restaurant Analysis project, several challenges were encountered while processing and analyzing the dataset.

- The original dataset contained missing values in important columns such as Rate, Restaurant Type, Cuisines, and Approximate Cost, which required careful data cleaning before analysis.
- Duplicate records were identified in the dataset and removed to improve data quality and ensure accurate results.
- The Approximate Cost column contained commas and was stored as text, so it had to be cleaned and converted into a numeric data type for calculations.
- Restaurant ratings were stored in different formats (such as "4.1/5"), requiring preprocessing to extract the numeric values.
- Importing the cleaned dataset into PostgreSQL required creating the appropriate table structure and ensuring data types matched correctly.
- Designing a professional Power BI dashboard involved selecting meaningful KPIs, appropriate visualizations, and maintaining a clean and user-friendly layout.
- Selecting the most relevant SQL queries from many possible analyses required focusing on business-oriented questions that provide actionable insights.

Despite these challenges, each issue was successfully resolved using Python (Pandas), PostgreSQL, and Power BI, resulting in a clean dataset and a professional analytical dashboard.

Conclusion

The Bangalore Zomato Restaurant Analysis project successfully transformed raw restaurant data into meaningful business insights using Python, PostgreSQL, and Power BI.

Python was used to clean, preprocess, and prepare the dataset by handling missing values, removing duplicate records, converting data types, and exporting a clean dataset for further analysis. PostgreSQL was then used to perform business-oriented SQL queries that helped identify popular restaurant locations, top cuisines, restaurant types, customer preferences, average ratings, pricing trends, online ordering patterns, and table booking statistics. Finally, Power BI was used to create an interactive dashboard that presents the findings through KPIs, charts, and slicers, making the analysis easy to understand and explore.

The analysis revealed that locations such as BTM, Whitefield, and HSR have the highest concentration of restaurants. North Indian cuisine and Casual Dining restaurants dominate the Bangalore food market, while online ordering is supported by a majority of restaurants. The dashboard also highlights customer voting patterns, average ratings, and pricing trends that can help restaurant owners make informed business decisions.

This project strengthened practical skills in data cleaning, SQL querying, data visualization, dashboard design, and business analysis. It demonstrates the complete workflow of a Data Analyst—from collecting and preparing data to generating actionable insights and presenting them through interactive dashboards. The knowledge and experience gained from this project can be applied to real-world business intelligence and analytics projects across different industries.

References

- [Kaggle – Zomato Bangalore Dataset](#)
- [Pandas Documentation](#)
- [PostgreSQL Documentation](#)
- [Power BI Documentation](#)